A Firm-Level Microsimulation for VAT Policy Analysis

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The problem: VAT as a tax *notch*

The UK VAT registration threshold is a notch, not a kink: once turnover crosses the threshold, 20% VAT is due on the firm's *whole* turnover, not just the excess.

- Among the highest thresholds in the OECD, sitting in a dense part of the firm-size distribution.
- Frozen at £85,000 from Apr 2017 to Mar 2024 — the longest nominal freeze in the tax's history — then raised to £90,000 in Apr 2024.
- Fiscal drag pushes ever more firms toward the threshold; firms bunch below it and slow their growth as they approach.

Contribution

An open, reproducible firm-level microsimulation for costing VAT threshold reforms, released with its synthetic-data generator. It prices three families of reform on a *common static basis*:

- Level — move the threshold up or down.
- Shape — a graduated taper that phases VAT in.
- Rate — a reduced-rate band above the threshold.

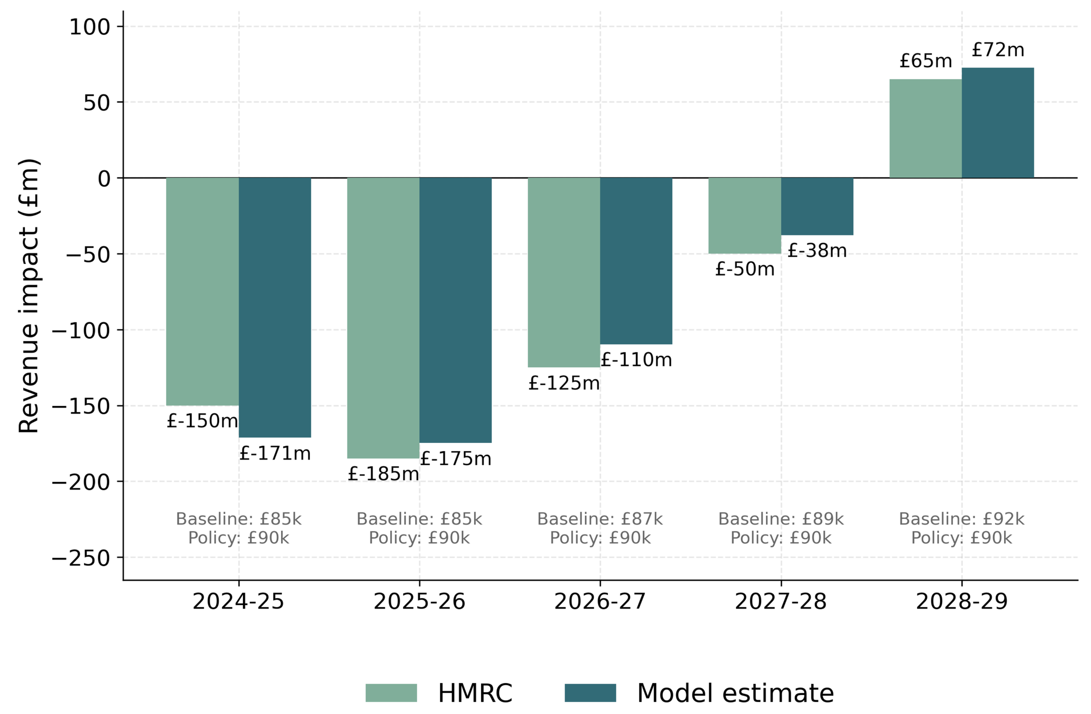
Shape reforms have no single point of excess mass, so reduced-form bunching methods cannot evaluate them — a structural model can.

Method

- Synthetic micro-data: draw published ONS firm counts by sector × turnover band; assign inputs, value added $v = y - x$, and employment. The threshold is a generator parameter, so data can be regenerated under any counterfactual.
- Calibration: firm weights $w_i = e^{\theta_i}$ fit by gradient descent to HMRC counts and net VAT liability by band and sector, and ONS employment totals.
- Result: 2.94m records weighted to ≈2.5m UK firms at ≈90% calibration accuracy.
- Static cost: $R(T) = \sum_i w_i v_i \mathbb{1}\{y_i \geq T\}$ — revenue moves only via firms whose registration status flips.

Validation against HMRC

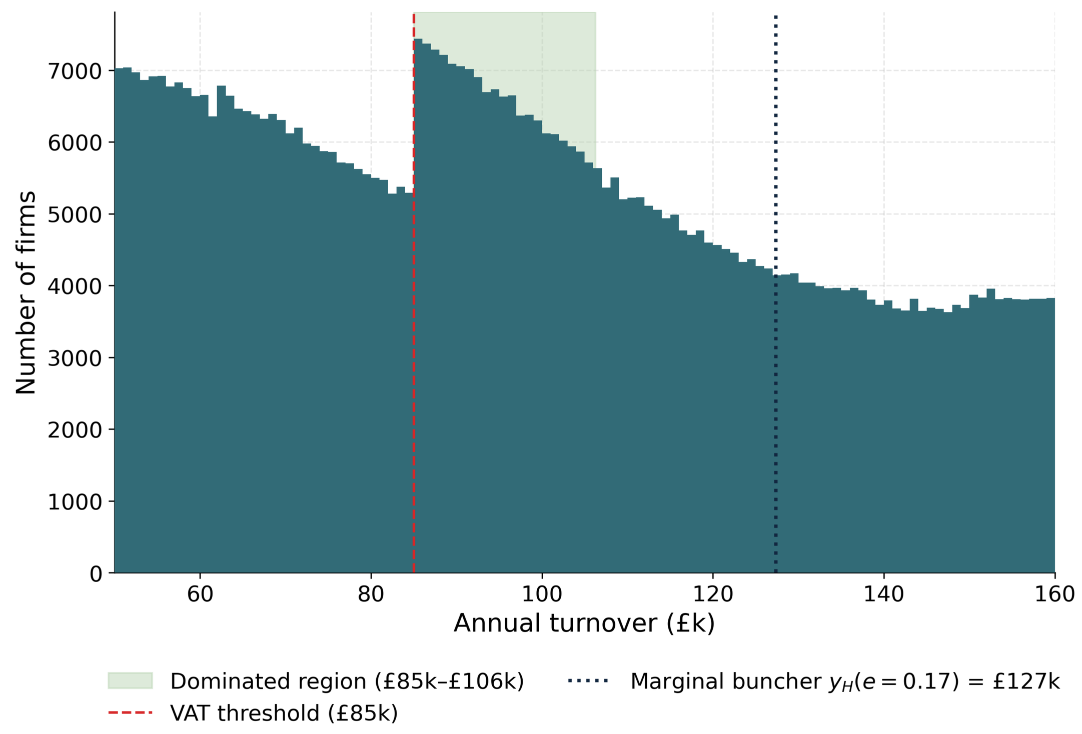
Costing the £85k→£90k (Apr 2024) increase gives −£175m in 2025–26 vs the Government's published −£185m — agreement within £10m, sharing the same turning point (a small gain by 2028–29).



The notch and its *dominated region*

The notch's distortion is characterised exactly by the dominated region: the band above the threshold, $a = T^* \frac{\tau}{1-\tau}$, in which no firm can profitably locate. It is elasticity-free.

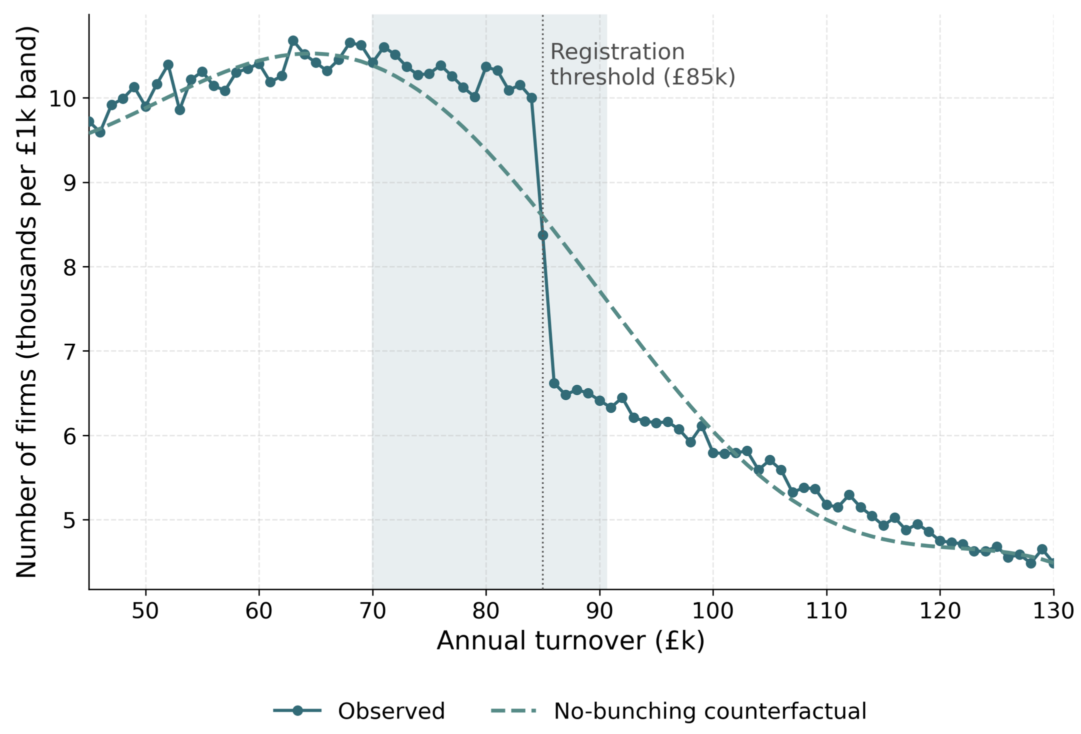
- Width a =£21,250 (£85,000–£106,250) — an order of magnitude wider than a kink's ∼£1k.
- Spans ≈137,000 weighted firms; notch jump at the threshold $= \tau T^* =$ £17,000.



Bunching here is *mechanical*

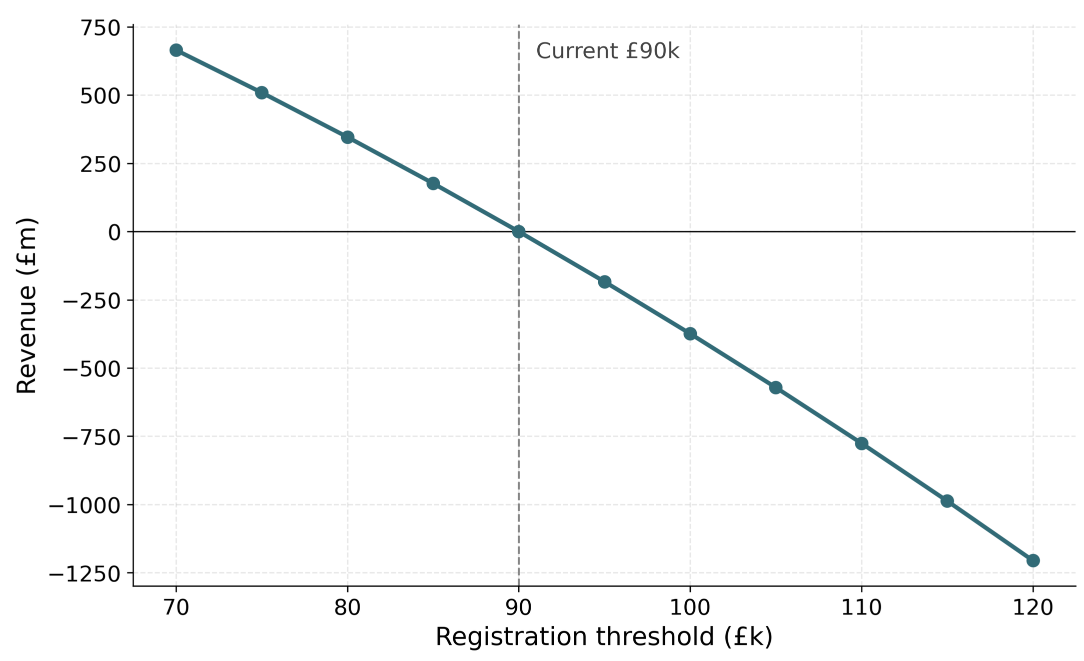
The estimated density step is a calibration artefact, not behaviour — a transparency lesson for synthetic data.

- Estimator: bunching ratio $b = 0.060$, excess mass 8,712 firms.
- Placebo (remove the band-step target): collapses to ≈99 firms — the step was inherited from coarse HMRC bands.



Reform menu (common static base, £183.6bn)

Reform	Static	Dominated region
Raise threshold to £100k	−£508m	Relocates band; removes none
Graduated taper £85k–£105k	−£336m	Removed entirely
Reduced rate 10% £85k–£105k	−£343m	Splits band (£22,569)
Reduced rate 15% £85k–£105k	−£171m	Splits band (£21,562)



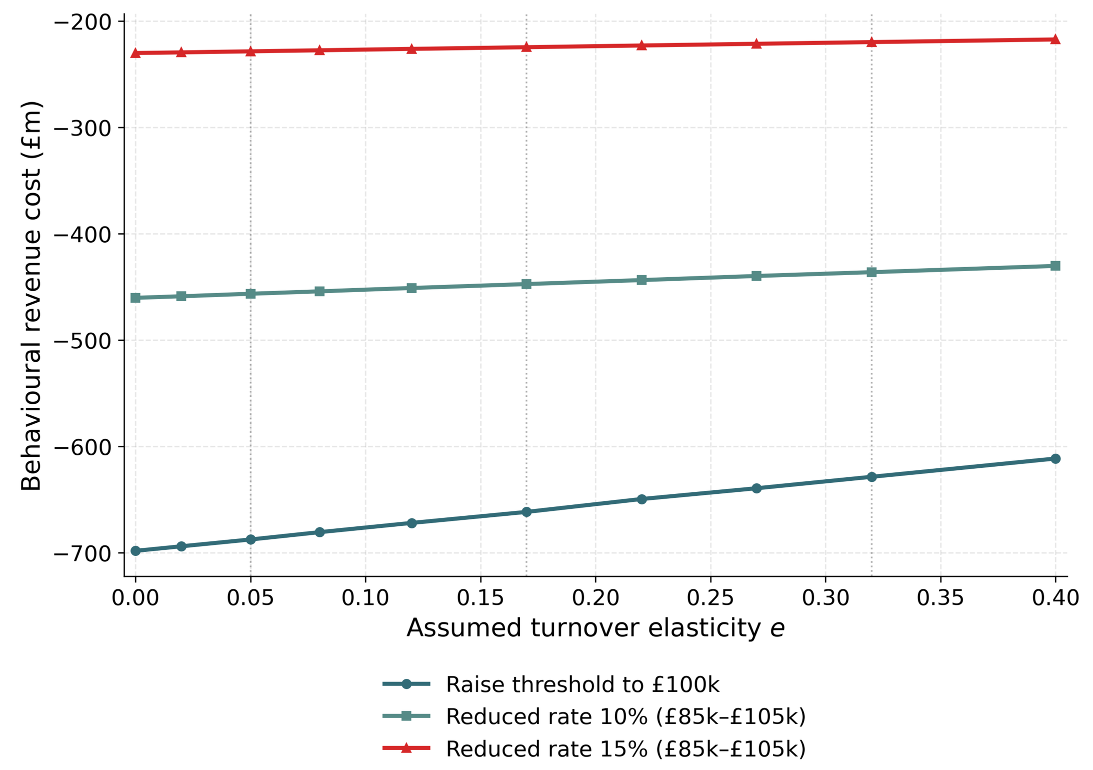
Headline result

The three reforms are not interchangeable:

- Raising the threshold is the *dearest* and removes *no* distortion.
- A reduced rate merely *splits* the dominated band in two.
- Only the graduated taper removes the distortion entirely — and for less (−£336m).

Behavioural sensitivity

Reforms are also priced behaviourally for an assumed turnover elasticity e, swept over $\{0.05, 0.17, 0.32\}$. Because e is assumed, not identified, this is a disciplined sensitivity range that nests the static cost as $e \to 0$.



At $e = 0.17$: raise-to-£100k −£292m (50,155 firms re-optimise); reduced-rate 10% −£273m; 15% −£135m.

Policy conclusions

- The static cost and the exact dominated region are the robust, elasticity-free anchors for comparing reforms.
- Behavioural magnitudes are a sensitivity band conditional on an unidentified e — not point estimates.
- Synthetic bunching must not be over-read as behaviour.

Scope & next steps

Taxes *turnover* (one input): a turnover-tax-notch approximation for consumer-facing, limited-reclaim firms. Abstracts from input-VAT reclaim and voluntary registration. Chief next task: identify e from administrative micro-data to turn the range into a point estimate, plus a welfare / MVPF analysis.

Code & contact

Open microsimulation and synthetic-data generator.

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